

Problem-Solution Fit

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Disaster Management Authorities - Government Agencies - Residents in Flood-Prone Areas	6. CUSTOMER LIMITATIONS FG. BUDGET, DEVICES [CL] - Limited access to real-time data - Lack of technical knowledge - Internet connectivity issues	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: - Weather forecasting services - Government warning systems Cons: - Low prediction accuracy - Late alerts - Limited local-level prediction
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty predicting floods accurately - Delayed disaster response - Frequent property and crop damage	9. PROBLEM ROOT / CAUSE [RC] - Climate change - Heavy rainfall - Poor drainage systems - Lack of accurate flood prediction	7. BEHAVIOR + ITS INTENSITY [BE] - Regularly checks weather updates - Follows emergency alerts - Takes preventive actions during heavy rainfall
3. TRIGGERS TO ACT [TR] - Heavy rainfall forecasts - Rising river water levels - Extreme weather alerts	10. YOUR SOLUTION [SL] Develop an AI-based Flood Prediction System that analyzes weather and rainfall data using Machine Learning to provide early flood predictions and timely alerts for better disaster preparedness.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: - Mobile App - Website - SMS Alerts
4. EMOTIONS BEFORE / AFTER [EM] Before: Fear, Uncertainty, Stress After: Confidence, Preparedness, Safety		OFFLINE: - Local Authorities - Television News - Radio Announcements